

COMP251 Introduction to Database Systems



LOYOLA
UNIVERSITY CHICAGO

INSTRUCTOR INFORMATION

Instructor Name: Christopher Fulton

Instructor Email: cfulton@luc.edu

Online Office Hours: Appointments are available upon request to allow greater flexibility. Please email me if you would like to schedule an individual session. The last 30 minutes of each class session will be used to address questions, ideas, and concerns.

COURSE INFORMATION

Course Start Date: 10/17/2026

Course End Date: 12/12/2026

Course Meeting Times: Monday 6 pm – 8:30 pm

Course Location: This is an 8-week, fully online synchronous course via Zoom. The Zoom link can be found in Brightspace.

Course Website: Brightspace is the learning management system where we will host all course materials. All lecture slides, assignment write-ups, course content, and learning material will be posted in Brightspace. The link below will take you to this semester's course site.

<https://brightspace.luc.edu/d2l/home/18346>

Course Description and Learning Outcomes:

This course explores ways in which data collections are organized, stored, analyzed, and manipulated. Topics include relational databases, the SQL query language, and the basics of XML and web interfaces to data sets. Applications from a variety of domains illustrate the course's key concepts.

Outcomes: Students will organize data in ways to emphasize relationships, write simple programs to process, visualize and graphically display data, mine data for patterns, and design web interfaces to data.

Course Objectives:

Upon successful completion of this course, students will be able to:

- I. Organize data in ways to emphasize relationships
- II. Write simple programs to process, visualize and graphically display data
- III. Mine data for patterns
- IV. Design interfaces to data
- V. Differentiate between use cases for Relational and NoSQL databases
- VI. Perform CRUD operations from both relational and NoSQL databases
- VII. Explain Big Data and Data Analytics

PRE/COREQUISITE COURSE(S)

Prerequisite: COMP 125 or COMP 150 or COMP 170 or COMP 180 or COMP/MATH 215

REQUIRED TEXTBOOK(S)

For this course, we will use a web-based textbook called ZyBooks. Please create an account and purchase the book at your earliest convenience (preferably before our first class session). Instructions on acquiring the book are listed below.

1. Sign in or create an account at learn.zybooks.com
2. Enter zyBook code: **LUCCOMP251FultonFall2026**
3. Subscribe

A subscription is **\$69**. Students may begin subscribing on Oct 03, 2026, and the cutoff to subscribe is Dec 09, 2026. Subscriptions will last until Dec 26, 2026.

OPTIONAL TEXTBOOK(S)

MongoDB Data Modeling and Schema Design, Author(s): S. Hoberman, D. Coupal, P. Desmarets
ISBN: 978-1634621984

LEARNER SUPPLIES

Below is a list of items you'll need for our course:

- **GitHub Account** - Your GitHub Account will host all your assignments and labs for the course. During the first week of class, we will walk through how to create a GitHub account and publish your work.
- **Internet Access and Google Chrome browser** - Access to high-speed internet is critical and essential for our weekly online synchronous sessions. In addition to internet access, you will need to have the Google chrome browser installed. You can find more information about the browser installation at the link below
 - <https://www.google.com/chrome/downloads/>
- **Reliable Computer and Webcam** - Due to our course being online and technical in nature, it is expected that your computer will have the specifications to handle installing software and the ability to utilize a webcam.
 - <https://www.luc.edu/its/services/desktoplaptopmobiledevices/technologypurchases/personaltechnologypurchases/personalpurchase recommendations/>

GRADING SCALE

Letter Grade/Percentage Equivalents for SCPS:

A 100% - 95%	B 86% - 88%	C 77% - 79%	D 65% - 70%
A- 92% - 94%	B- 83% - 85%	C- 74% - 76%	F <65%
B+ 89% - 91%	C+ 80% - 82%	D+ 71% - 73%	

GRADING RATIONALE

- Assignments = 30% of final grade | 3 Assignments (30 points each)
- Quizzes/Participation = 20% | 6 Quizzes (10 points each)

- ZyBook Reading = 13.33% | 8 Assigned Readings (5 points each)
- Lab Activities = 20% | 4 Lab Activities (15 total points)
- Exams = 16.66% | 1 Final Exam (50 Points)

Total Possible Points: 300

FORMATIVE ASSESSMENT

Assignments, Labs and ZyBook and instructor-assigned readings

SUMMATIVE ASSESSMENT

Quizzes, Final Exam

SCHEDULE

Session	Course Outcome(s)	Topics/Activities *All Zybook readings are due at 6pm prior to weekly session	Assessment(s)
Week 1	CO I	Introductions and Syllabus Review Download and Install MySQL Server and Workbench Understanding Data ZyBooks Reading: 1.1 – 1.4 Due (10/25)	Lab 1 – Released Lab 1: MySQL Workbench Due (10/25)
Week 2	CO I, CO IV	Database Design I – Conceptual Data Model ZyBooks Reading: 2.1 – 2.4 Due (10/26)	Assignment 1 Released Lab 2 - Released
Week 3	CO I, CO IV	Database Design II – Logical Data Model ZyBook Reading: 3.1 – 3.4 Due (11/2)	Lab 2: Conceptual Modeling using Draw.io – Due (11/11)
Week 4	CO I	Normalization ZyBook Reading: 4.1 – 4.3 Due (11/9)	Assignment 2 Released Lab 3 Released Assignment 1: ER Model Due (11/15)
Week 5	CO II, III, VI	SQL – Physical Data Model Working with MySQL Workbench ZyBook Reading: 5.1 – 5.13 Due (11/16)	
Week 6	CO II, III, VI	Advanced SQL ZyBook Reading: 6.1 - 6.8 Due (11/23)	Assignment 3 Released Lab 4 Released Lab 3: Basic Database Queries Due (11/29)

Week 7	CO V, VII	Intro. To Big Data NoSQL Databases JSON/XML Review ZyBook Reading: 7.1 – 7.6 Due (12/30)	Lab 4: JSON and XML Conversion Due (12/7) Assignment 2: Physical Model + Queries Due (11/29)
Week 8	CO VI	NoSQL Databases Cont. Brief Intro. to Database Architecture ZyBook Reading: 8.1 – 8.4 Due (12/7)	Assignment 3: NoSQL MongoDB Due (12/11) Final Exam

ATTENDANCE

Attendance and active participation are required for all synchronous class sessions. Required sessions are listed below.

- October 19, 2026, 6 pm – 8:30 pm
- October 26, 2026, 6 pm – 8:30 pm
- November 2, 2026, 6 pm – 8:30 pm
- November 9, 2026 6 pm – 8:30 pm
- November 16, 2026, 6 pm – 8:30 pm
- November 23, 2026, 6 pm – 8:30 pm
- November 30, 2026, 6 pm – 8:30 pm
- December 7, 2026, 6 pm – 8:30 pm

SUCCEEDING IN THIS CLASS

As the instructor, my job in this class is to organize the material coherently, give helpful lectures, provide a framework that combines enough challenge and support for success, and grade reasonably. A reasonable academic expectation is that you spend 2 hours outside of class for every hour spent in class. For a few of your assignments, you may go over this time estimate.

- Complete the assigned reading and review material before each class session.
- Take notes and participate in all the learning activities found in the slide presentations
- Complete labs and assignments to the best of your ability

LATE WORK POLICY

All assignments and labs must be submitted by the due date and time listed in Brightspace. If the assignment is not turned in by the due date and time listed, the program will not receive points.

Students can receive up to 2 consecutive days beyond the original due date if the instructor is notified of an extension before the due date listed on the program. Once the extension expires and the assignment is not submitted, it will automatically receive a zero if it is not submitted within 2 consecutive days beyond the original due date. All quizzes must be submitted by the due date and time listed in Brightspace.

No exceptions will be made for quizzes and exams unless explicitly stated otherwise.

ACADEMIC INTEGRITY

Cheating, plagiarism, and any other forms of academic dishonesty will not be tolerated.

This course will follow the standards outlined by the institution. The link below outlines each standard and

sanctions for academic misconduct. Once you navigate to the webpage below, click on the “Academic Integrity” tab to review standards and sanctions.

<https://catalog.luc.edu/academic-standards-regulations/undergraduate/>

Any student who fails to follow the academic integrity policy on any coursework will receive a grade of 0.

PLAGIARISM

Work submitted that has been copied from another source, such as a book, periodical, website, or another student, and is not cited correctly, is considered plagiarism. Paraphrasing (rewording of text) or quoting any part from a source must be cited appropriately.

ACCOMMODATIONS

Loyola University Chicago provides reasonable accommodations for students with disabilities. Any student requesting accommodations related to a disability or other condition is required to register with the Student Accessibility Center (SAC). Professors will receive an accommodation notification from SAC, preferably within the first two weeks of class.

Students are encouraged to meet with their professor individually in order to discuss their accommodations. All information will remain confidential.

Please note that in this class, software may be used to audio record class lectures in order to provide equal access to students with disabilities. Students approved for this accommodation use recordings for their personal study only and recordings may not be shared with other people or used in any way against the faculty member, other lecturers, or students whose classroom comments are recorded as part of the class activity. Recordings are deleted at the end of the semester.

For more information about registering with SAC or questions about accommodations, please contact SAC at 773-508-3700 or SAC@luc.edu.

LEARNER SUPPORT

Tech Support HelpDesk: helpdesk@luc.edu 773-508-4487
Student Accessibility Center: <http://www.luc.edu/sac/>

DISCLAIMER

This syllabus is subject to change at the instructor's discretion with prior student notification.

Syllabus Updates:

10/4/2026

- Initial release of Fall 2026 Syllabus